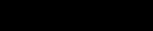


AKHIL JAINI

Data Scientist | Machine Learning Engineer

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SUMMARY

Experienced Computational and Data Scientist with hands-on experience designing real-time, high-performance data systems, predictive statistical models, and applied machine learning solutions for large-scale, noisy datasets. Bringing expertise in advanced Python and SQL analytics, and scientific computing. Proven track record of translating complex analytical problems into production-grade software, delivering measurable performance gains, and communicating technical findings to both technical and non-technical stakeholders. Currently completing a PhD focused on real-time anomaly detection and statistical experimentation at extreme data scale.

TECHNICAL SKILLS

Programming: Python, SQL, C, C++, CUDA C, Rust, JavaScript, TypeScript

Machine Learning: PyTorch, Scikit-Learn, TensorFlow, Classification, Regression, XGBoost, Random Forest, RNNs, Clustering, Statistical Modelling, Hypothesis Testing, A/B Testing, Feature Engineering

LLM & GenAI Systems: Multi-Agent Systems (LangGraph), Retrieval-Augmented Generation (RAG), LLM Evaluation (vLLM, Ollama), Parameter-Efficient Fine-Tuning (LoRA), Model Quantisation, Prompt & Context Engineering

Engineering Practices: FastAPI, REST APIs, Docker, Git, GitHub Actions, Automated Testing, CI/CD

Data & Infrastructure: PostgreSQL, SQLite, Linux, AWS, Distributed/HPC Computing (OzStar/Slurm), Data Visualisation, EDA

EXPERIENCE

PhD Researcher (Real-Time Data Systems) – Swinburne University of Technology, Melbourne

June 2023–Present

- Engineered predictive feature-matching and spatial indexing algorithms across multi-terabyte datasets in Python and SQL, improving localisation precision by 4x (from ~1 to <0.25 arcseconds) and enabling reliable downstream analysis at a precision not previously achievable.
- Built and deployed a real-time streaming pipeline processing radio telescope sensor data at >20TB/s throughput, implementing anomaly-detection algorithms that flag rare, millisecond-scale events with >99% reliability.
- Developed statistical and time-series models to characterise periodicity and signal structure in noisy, high-dimensional data, directly informing follow-up data-collection strategy with a national research telescope.
- Led a multidisciplinary international project and team to analyse and characterise rare transient signals from multiple astronomical instruments; co-led a parallel multi-telescope data collection campaign to validate findings for polarisation and time-series analyses.
- Co-authored 12+ peer-reviewed publications in high-impact astrophysics journals (3 as first author) and presented findings at 15+ invited talks and international conferences, translating complex results into clear, actionable takeaways for technical and non-technical audiences.
- Collaborated with international, cross-functional technical teams to convert research-grade analysis into production-grade software adopted by multiple external organisations.

Research Software Engineer – Indian Institute of Astrophysics, Bangalore

August 2021–April 2023

- Designed and validated automated data-quality workflows to detect and correct systematic errors in high-precision instrument data, improving measurement reliability across a multi-year dataset.
- Built Python and C++ control interfaces and automated data-extraction pipelines for real-time instrument calibration and control, high-precision spectrophotometric data acquisition, and telemetry monitoring on a state-of-the-art space-based ultraviolet telescope.
- Designed and simulated optical system components for a next-generation imaging instrument, validating performance under rigorous laboratory test conditions.

Computational Research Assistant – Raman Research Institute, Bangalore

January 2018–March 2020

- Designed and implemented a full-scale Python simulation framework modelling multi-variable sensor networks for an orbiting radio interferometer, performing predictive risk analysis and scenario modelling under the rigorous operating conditions of outer space.

Software Engineering Intern – National Centre for Radio Astrophysics, Pune

May 2017–July 2017

- Built a GPU-accelerated (CUDA C) real-time signal-processing pipeline for high-throughput event detection, porting and extending existing code and validating performance against simulated streaming data.

ML Engineering Intern – CSIR-Central Leather Research Institute, Chennai

May 2016–July 2016

- Applied unsupervised machine learning (K-means and hierarchical clustering) and advanced image-processing techniques in MATLAB and C++ to automatically detect surface defects and grade material quality, improving on conventional grading methods.

PROJECTS

Astronomical Candidate Classification Pipeline (*Scikit-learn, Keras, XGBoost, U-Net, Regression*)

- Built an end-to-end ML pipeline for detecting and characterising transient signals in noisy, high-volume data, combining Random Forest/XGBoost classification and engineered spatial-temporal features to filter noise and radio-frequency interference.
- Developed a lightweight 2D U-Net segmentation model to isolate signal patterns in time-frequency data, followed by automated curve fitting and signal processing to estimate key parameters and generate quantitative event measurements.

Bushfire Early Warning System (*PyTorch, SQLite, FastAPI, Streamlit, Docker, MLflow*)

- Built a Sentinel-1 SAR fire-detection pipeline that calibrates and speckle-filters raw radar imagery into a 6-channel feature stack (temporal backscatter deltas, polarisation ratio, GLCM texture), then trains a ResNet-34 U-Net with a Focal+Dice loss under spatial block cross-validation to handle severe class imbalance and prevent geographic data leakage.
- Served the model as a FastAPI microservice returning GeoJSON fire perimeters with a Streamlit + Folium dashboard, tracked experiments via MLflow, and containerised the stack with Docker and a GitHub Actions CI pipeline.

Agentic Document & Code Review Assistant (*LangGraph, Ollama, ChromaDB, Docker, Pydantic*)

- Architected a stateful multi-agent orchestration engine using LangGraph and local Ollama models (Llama 3.1 & Qwen 2.5-Coder), deploying a cyclic supervisor pattern with memory checkpointing to autonomously audit codebases and query technical documentation.
- Engineered an asynchronous Dockerised FastAPI & WebSocket backend delivering real-time streaming execution, integrating custom Python AST static parsers, ChromaDB vector RAG, dynamic LLM hot-swapping, and strict Pydantic schema validation.

AstroConvert (*Flet, NumPy, SciPy, AstroPy*)

- Designed and developed a cross-platform desktop tool that intelligently detects, validates, and transforms heterogeneous scientific coordinate formats, substantially reducing manual preprocessing overhead through automated input recognition.

Apex Drift: Neuroevolution-Based Autonomous Racing Platform (*PyTorch, Pygame, Deep Learning, Optimisation*)

- Developed an autonomous driving system using neuroevolution, evolving neural-network populations through selection, crossover and mutation to learn driving behaviours without supervised training data.
- Engineered a physics-based simulation environment with raycast perception, non-linear vehicle dynamics and multi-objective fitness optimisation, supported by automated testing and model versioning for reproducible evaluation.

Event Detection & Key Actor Identification in Multi-Person Sports Videos (*PyTorch, YOLOv8, OpenCV, Gradio, Docker, GCP*)

- Modernised a computer vision pipeline for multi-person sports video, combining YOLOv8 object detection, Hungarian-algorithm multi-object tracking and a dual-branch BiLSTM with temporal attention for event classification and key-actor localisation.
- Achieved 99.09% event-classification accuracy on 110 held-out broadcast clips across basketball, cricket and football, and deployed the end-to-end system as a Dockerised Gradio application on Google Cloud Run with automated testing.

Diffusion Model Fine-Tuning (*Stable Diffusion XL, LoRA, Fooocus*)

- Fine-tuned a parameter-efficient LoRA adapter for a Stable Diffusion XL foundation model, curating and preprocessing a custom image dataset with automated and manual captioning to ensure concept fidelity without overfitting.
- Configured network rank, alpha, and optimiser hyperparameters to fine-tune U-Net and text-encoder components, and systematically evaluated intermediate checkpoints against baseline outputs using Fooocus to balance prompt adherence against generative flexibility.

Local LLM Inference Benchmarking (*vLLM, Ollama*)

- Built a benchmarking framework comparing local LLM inference engines (vLLM vs. Ollama) across open-weight models including Llama 3.1, Qwen 3-Coder, Qwen 3.8, Gemma 4, and DeepSeek V4-Flash, measuring time-to-first-token, token throughput, and memory utilisation across varying concurrency and context-window settings.
- Evaluated quantisation trade-offs and dynamic batching strategies to compare local serving configurations.

Full list of Projects: <https://github.com/jainiakhil>

RESEARCH IMPACT

First author of multiple peer-reviewed publications in top-tier international astrophysics journals, developing scalable computational methods, data processing algorithms, and software adopted across international astronomy collaborations.

Full list of Publications: ui.adsabs.harvard.edu/public-libraries/6AWkqcbzTQWau_usyI5V8Q

EDUCATION

PhD in Computational Astrophysics Swinburne University of Technology, Melbourne, Australia	2023-Ongoing
Master of Technology in Astronomical Instrumentation Indian Institute of Astrophysics, Bangalore, India and University of Calcutta, Kolkata, India	2020-2022
Bachelor of Engineering (Honours) in Electrical and Electronics Birla Institute of Technology and Science – Pilani, Pilani, India	2014-2018